21/20 Bzalo keepsitup

Blessed G/Michael Catholic General Secondary School

Mathematics Test one 2018 E. C for grade 10th

Name: Bizuayehu Wondimagegnehu Class: 10th B **Time allowed:- 1.10'**

No = 12

I. Choose the best answer among the given alternatives.

D---1. Which one of the following is not true?

A. The produce of two polynomial function is always a polynomial
B. The difference of two polynomial function is always a polynomial
C. The sum of two polynomial function is always a polynomial
D. The quotient of two polynomial function is always a polynomial

C---2. Which of the following is true about $f(x) = 6x^2 - x - 1$

A. The function $f(x) = 6x^2-x-1$ has no x – intercepts
B. f has y intercept of (0, 1)
C. The graph of f is an upward parabola
D. The vertex of is (12, -24)

10

B---3. When the function $f(x) = 2x^3+4x^2 - 5x+2$ is divided by x+1, the remainder is ----

A. 5 B. 9 C. 12 D. 16

A---4. When $kx^2 - 5x+6$ is divided by (x -1), the remainder is 4 what is the value of k?

A. 3 B. 5 C. 8 D. 11

A---5. If (x+2) and (x-2) are the factors of the polynomial $3x^3 + mx^2 - Nx+20$ then find the value of M

A. -5 B. 12 C. 4 D. 3

B---6. In the above question number 5 what is the value of N

A. 3 B. 12 C. 16 D. 24

B---7. The polynomial $x^3 + Ax^2+Bx - 12$ is divisible by x+3 and gives remainder (-6) when divided by x+1 what is the value of A?

A. 2 B. 3 C. 4 D. 5

C---8. What is the value of B in the above question?

A. 3 B. 4 C. -4 D. 5

D---9. If 2x+1 is the factor of $2x^2+7x+3$ then what is the other factor

A. x+2 B. x - 2 C. x - 3 D. x + 3

D---10. What is the minimum values of $f(x) = 2x^2+7x+3$

A. $\left(-7/4, -25/8\right)$ B. $\left(-7, -25/4\right)$ C. (-1.75) D. -3.125

II. Give short answer only

11. What is the domain of $f(x) = \sqrt{8-2x}$ = domain : $x \leq 4$

12. What is the x intercept of $f(x) = x^2-7x+12$ {3, 4}

13. What is the vertex of the graphs of $g(x) = -x^2+6x-8$? = (3, 1)

14. What is the slope of the line with equation $y = 3x+9$ = 3

15. What is the inverse of the relation $R = \{(x,y) 4y = 2x+6\}$?

$4x = 2y+6$

$\frac{2y}{2} = \frac{4x-6}{2}$ $y = 2x-3$

III. Work out (Show the necessary steps)

16. Draw the graph of the relation $R = \{(x,y) : y \leq x+3, x \leq 2$ and $y \geq -2x-6\}$ and Find the domain and Range of R

16, $y \leq x+3$ y-intercept = (0, 3) Shaded downward
x-intercept : (-3, 0)

$x \leq 2$ y-intercept : (0, 0) Shaded to left
x-intercept : (2, 0)

$y \geq -2x-6$ y-intercept : (0, -6) Shaded upward
x-intercept : (-3, 0)

17. Given a quadratic function $f(x) = -x^2-8x-7$

a. what is the vertex of the function

b. what are the x intercepts

c. what is the y – intercept

d. draw its graph

17, Vertex = $\left(\frac{-b}{2a}, \frac{4ac-b^2}{4a}\right)$ & $a = -1, b = -8, c = -7$

$= \frac{-(-8)}{2(-1)}, \frac{4(-1)(-7)-(-8)^2}{4(-1)}$

$= (-4, 9)$

x-intercept = $\frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$= \frac{(-8) \pm \sqrt{(-8)^2-4(-1)(-7)}}{2(-1)}$

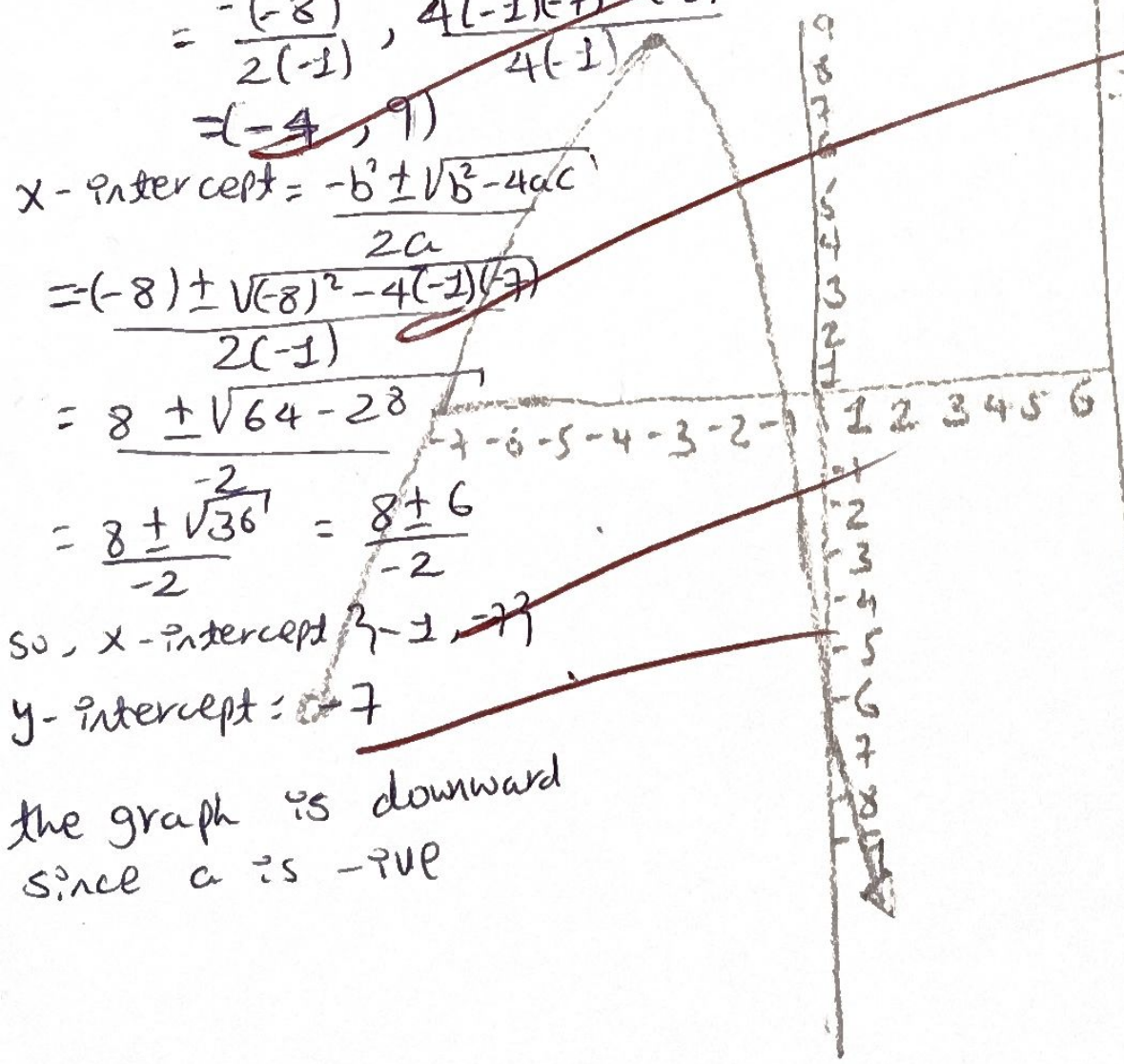
$= \frac{8 \pm \sqrt{64-28}}{-2}$

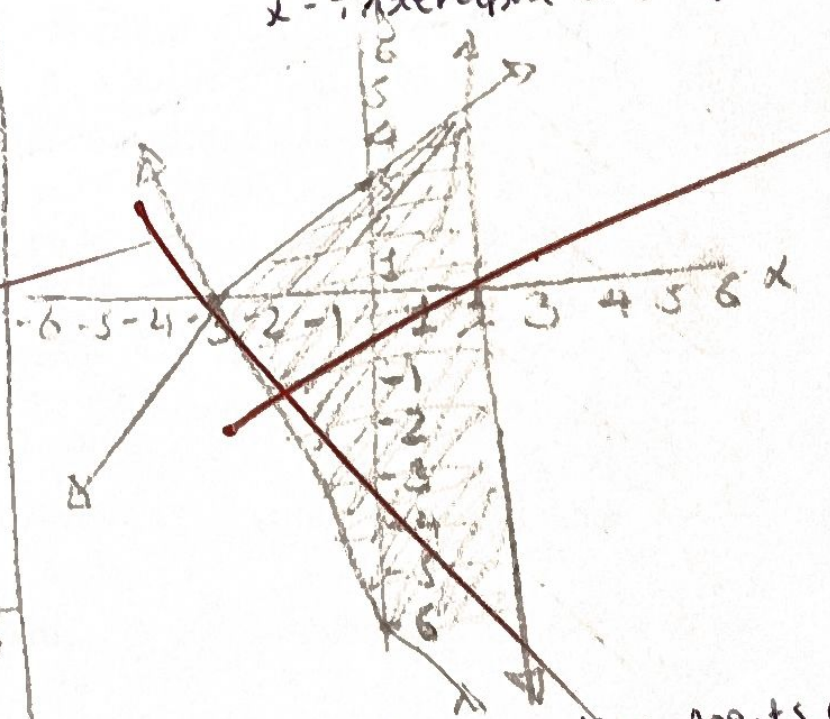
$= \frac{8 \pm \sqrt{36}}{-2} = \frac{8 \pm 6}{-2}$

so, x-intercept {-1, -7}

y-intercept : -7

the graph is downward since a is -ve





Since their intersection points are $(\frac{2}{5}, 2)$ & $(-3, -10)$ due to $x = 2$ we substitute it on the two eqn's get $5 \leq -10$ so the

domain = ~~$x \leq 2$ & $x \geq -3$~~

range = ~~$y \leq 5$ & $y \geq -10$~~